

- There are 6 Questions for 60 marks. Each part is for 2 marks.
  - Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class. You may use any result from class directly (without re-deriving the proof). Clearly justify any statement you make, even if the problem doesn't explicitly ask you to.
  - Good luck!
- Consider the vector valued function  $f_A : \mathbb{R}^3 \mapsto \mathbb{R}^3$  given by  $f_A(\underline{x}) = A\underline{x}$ , where  $A$  is a  $3 \times 3$  matrix.
    - What are the Jacobian, curl  $\nabla \times f_A$  and divergence  $\nabla \cdot f_A$  of  $f_A$ ?
    - What is the flux of  $f_A$  through the unit sphere centred at the origin?
    - Find the flux through the cylinder  $\{(x, y, z) : x^2 + y^2 \leq 1, -1 \leq z \leq 1\}$ .
    - What is the line integral  $\oint_C f_A \cdot d\underline{x}$  on the curve  $C = \{(x, y, 0) : x^2 + y^2 = 1\}$  oriented counter-clockwise?
    - For what matrices  $A$  is the vector field  $f_A$  conservative?
    - Can you write  $A = B + C$  as a sum of matrices such that  $\nabla \cdot f_A = \nabla \cdot f_B$  and  $\nabla \times f_A = \nabla \times f_C$ ?
  - What is the area of the region  $T_2 \subseteq \mathbb{R}^2$  given by  $T_2 = \{(x, y) : x + y \leq 1, x \geq 0, y \geq 0\}$ ?
    - What is the volume of the tetrahedron  $T_3 \subseteq \mathbb{R}^3$  given by  $T_3 = \{(x, y, z) : x + y + z \leq 1, x \geq 0, y \geq 0, z \geq 0\}$ ?
    - What is the volume of the skewed tetrahedron  $S_3 \subseteq \mathbb{R}^3$  given as  $S_3 = \{(u, v, w) : u + v + w \leq 1, u \geq -v, v \geq -w, w \geq -v\}$ ? *Hint: You may consider making a change of variable.*
    - What are the maximum and minimum values of  $2x + 5y + 7z$  on the tetrahedron  $T_3$ ?
    - What is the minimum value of  $4x^2 + 8y^2 - 8xy - x$  on the region  $T_2$ ?
  - Consider the following scalar valued function  $f : \mathbb{R}^2 \mapsto \mathbb{R}$

$$f(x_1, x_2) = (a_{11}x_1 + a_{12}x_2 - b_1)^2 + (a_{21}x_1 + a_{22}x_2 - b_2)^2,$$

for real constants  $a_{ij}$  and  $b_i$ .

- What is the gradient of  $f$ ? Give your answer in 'simplified' matrix form, if possible. You can consider the  $2 \times 2$  matrix  $A$  as

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

- What are the local min, local max and saddle points of  $f$ ?
- At the origin  $(x_1 = 0, x_2 = 0)$ , which direction does the function  $f$  decrease the most?
- Suppose the vector  $(x_1, x_2)$  changes by an amount (magnitude)  $\epsilon$  from the origin along the direction you mentioned in part (c). For small values of  $\epsilon$ , what is the approximate change in  $f$ ?

Now, suppose  $x_1 = r \cos \theta$  and  $x_2 = r \sin \theta$  are written in polar coordinates.

- What is the gradient of  $g(r, \theta) = f(r \cos \theta, r \sin \theta)$ ? You may leave your answer in matrix form.

- Consider the function  $f(x)$  defined as the series

$$f(x) = 1 + 2x + x^2 + 2x^3 + x^4 + 2x^5 + x^6 + \dots$$

What is the radius of convergence of  $f$ ? Find an explicit expression for  $f$ .

- Find the Taylor series for  $\cos x$ . Recall that  $e^x = 1 + x + x^2/2! + x^3/3! + \dots$

- (c) What is  $\lim_{x \rightarrow 0} (1 - \cos x) / x^2$
- (d) Find  $\cos 1$  up to the second decimal point.
- (e) Is the series  $\sum n^n / (2n)!$  convergent ?
5. Consider a robot with unit mass that can move on a frictionless (one-dimensional) line. We will apply a force at time  $t$  (along the line) given by  $f_j$  for  $j - 1 \leq t \leq j$  where  $j = 1, \dots, n$ , so that the applied force is constant over time intervals of length 1. So a force of  $f_1$  is applied for time  $0 \leq t \leq 1$ , force  $f_2$  for time  $1 \leq t \leq 2$  and so on.

In this problem, we assume that the robot is initially at the origin and at rest. We only operate the robot from time  $t = 0$  to  $t = N$ . We apply forces  $f_1, f_2, \dots, f_N$ ; which we denote by  $\underline{f}$ :

$$\underline{f} = (f_1 \quad f_2 \quad \dots \quad f_N)^T.$$

Let  $x_{\text{pos}}(t)$  be the position of the robot at time  $t$ . We denote by vector  $\underline{x}(t)$  the positions of the robot at times  $t_1, t_2, \dots, t_m$

$$\underline{t} = (t_1 \quad t_2 \quad \dots \quad t_m)^T, \quad \underline{x}(t) = (x_{\text{pos}}(t_1) \quad x_{\text{pos}}(t_2) \quad \dots \quad x_{\text{pos}}(t_m))^T.$$

For the case when  $t_i = i$ , we denote by  $\underline{x}_{1:N}$  the positions:

$$\underline{x}_{1:N} = (x_{\text{pos}}(1) \quad x_{\text{pos}}(2) \quad \dots \quad x_{\text{pos}}(N))^T$$

- (a) Give a recursive expression for  $x_{\text{pos}}(n)$  for  $n = 1, 2, \dots, N$ .
- (b) Express  $\underline{x}_{1:N} = A\underline{f}$  for a suitable matrix  $A$ . Is this relationship linear?
- (c) Comment on the structure of the matrix  $A$ . What is the rank of  $A$ ?
- (d) Find the position  $x_{\text{pos}}(t)$  at an arbitrary time  $t$ , where  $0 \leq t \leq N$ . Your answer should be a function of  $t$  and the vector of input forces  $\underline{f} \in \mathbb{R}^N$ . You may express the answer in terms of  $x_{\text{pos}}(n)$  from part (a).
- (e) Now consider  $m$  time-stamps  $t_1, t_2, \dots, t_m$  that are all between 0 and 1. Express  $\underline{x}(t) = B\underline{f}$  for a suitable matrix  $B$ . Comment on the structure and rank of  $B$ .
- (f) Let us generalize the above. Suppose we now have timestamps  $t_1, t_2, \dots, t_m$  between 0 and  $N$  (not necessarily between 0 and 1 as in previous part). If you express  $\underline{x}(t) = C\underline{f}$  for a suitable matrix  $C$ , what can you say about the rank of  $C$ ?

*Hint:* With initial position  $x_0$ , initial velocity  $v_0$  and constant acceleration  $a$ , the position and velocity at time  $t$  are given by  $x(t) = x_0 + v_0 t + \frac{1}{2} a t^2$ ,  $v(t) = v_0 + a t$ .

6. This problem continues from the previous problem.

Suppose there  $M$  milestones on the line and the goal is to design a sequence of input forces for the robot to be at these milestones at specific times: the robot needs to be at milestone 1 at time  $t_1$ , milestone 2 at time  $t_2$  and so on. The timestamps  $t_1, t_2, \dots, t_M$  (recorded in vector  $\underline{t}_{\text{target}}$ ), and positions of the milestones  $x_1, x_2, \dots, x_M$  (recorded in vector  $\underline{x}_{\text{target}}$ ) are given to us.

- (a) Give an algorithm such that with input  $\underline{x}_{\text{target}}$  and  $\underline{t}_{\text{target}}$ , the algorithm finds the forces  $\underline{f}$  to apply.
- (b) Could there be values of the milestones  $\underline{x}_{\text{target}}$  and timestamps  $\underline{t}_{\text{target}}$  for which it is impossible to find a sequence of forces ?
- (c) How many arithmetic operations (give your answer in Big Oh  $O()$  notation) does your algorithm cost?

Now suppose we allow for the robot to miss some of these milestones, so long as we are reasonably close to it. We formulate the following norm-minimization problem

$$\min_{\underline{f}} \|\underline{x}(\underline{t}_{\text{target}}) - \underline{x}_{\text{target}}\|_2$$

- (d) Give an algorithm to solve the above problem. How many arithmetic operations does the algorithm cost ?